Taller III

ítem 1

$x=0$

1) Det. ecuación recta tg a la curva $y^2e^{2x} = 3y + x^2$ en el Punto de abscisa cero que no corresPonde al origen

$$2y \cdot y' \cdot e^{2x} + y^2 \cdot e^{2x} \cdot 2 = 3y' + 2x$$
$$2yy'e^{2x} - 3y' = 2x - 2y^2e^{2x}$$
$$y'(2ye^{2x} - 3) = 2x - 2y^2e^{2x}$$
$$y' = \frac{2x - 2y^2e^{2x}}{2ye^{2x} - 3}$$

Con $x=0 \Rightarrow y^2 = 3y \Rightarrow y^2 - 3y = 0 \Rightarrow y(y-3) = 0 \Rightarrow y=0,\ \boxed{y=3}$

$$y' = \frac{2\cdot 0 - 2(3)^2e^0}{2\cdot 3\cdot e^0 - 3} \Rightarrow y' = \frac{-18}{3} \Rightarrow -6$$

Recta tg $\Rightarrow y - 3 = -6(x-0)$
$$y = -6x + 3$$

Halle $\frac{dy}{dx}$ si $y = (\operatorname{sen} x)^{\sqrt{x}}$

$$\frac{dy}{dx} = e^{\sqrt{x}\ln(\operatorname{sen}x)}$$
$$\frac{dy}{dx} = e^{\sqrt{x}\ln(\operatorname{sen}x)} \cdot \frac{1}{2\sqrt{x}} \cdot \ln(\operatorname{sen}x) + \sqrt{x} \cdot \frac{1}{\operatorname{sen}x} \cdot \cos x$$
$$\frac{dy}{dx} = \frac{e^{\sqrt{x}\ln(\operatorname{sen}x)} \cdot \ln(\operatorname{sen}x)}{2\sqrt{x}} + \frac{\sqrt{x}\cos x}{\operatorname{sen}x}$$

item III

1) $\lim\limits_{z \to 0} \dfrac{sen(2z) + 7z^2 - 2z}{z^2(z+1)^2} \overset{0/0}{\Rightarrow} \dfrac{\cos(2z)\cdot 2 + 14z - 2}{2z(z+1)^2 + z^2 \cdot 2(z+1)} \overset{0/0}{=} \dfrac{-2sen(2z)\cdot 2 + 14}{\underbrace{2(z+1)^2}_{2} \underbrace{\cancel{+ 2z\cdot 2(z+1)}}_{0} \underbrace{\cancel{+ 4z(z+1) + 2z^2}}_{0}} \Rightarrow \dfrac{14}{2} \Rightarrow 7$

2) $\lim\limits_{x \to 1/2} \dfrac{arcsen(x) - \frac{\pi}{6}}{2x - 1} \overset{0/0}{\Rightarrow} \dfrac{\frac{1}{\sqrt{1 - x^2}}}{2} \Rightarrow \dfrac{1}{2\sqrt{1 - x^2}} \Rightarrow \dfrac{1}{2\sqrt{\frac{3}{4}}} \Rightarrow \dfrac{1}{\sqrt{3}} \Rightarrow \dfrac{\sqrt{3}}{3}$

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item IV

1) $f(x) = 2x + \frac{1}{x} \Rightarrow \dfrac{2x^2 + 1}{x}$

a) Det. Pontos criticos (Rel o max)

$f'(x) \Rightarrow 2 - \frac{1}{x^2} = 0$, $x = \pm \frac{1}{\sqrt{2}}$

comcava si $f''(x)$ +

convexa si $f''(x)$ −

	$-\infty$	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	$+\infty$
$2 - \frac{1}{x^2}$	+	−	+	
	↗	↘	↗	

$2\cdot\left(-\frac{1}{\sqrt{2}}\right) + \dfrac{1}{\frac{-1}{\sqrt{2}}} = -2\sqrt{2}$

$2\cdot\left(\frac{1}{\sqrt{2}}\right) + \dfrac{1}{\frac{1}{\sqrt{2}}} \Rightarrow 2\sqrt{2}$

$f''(x) \Rightarrow \dfrac{2}{x^3}$, $f''\left(-\frac{1}{\sqrt{2}}\right) \Rightarrow \dfrac{2}{-\frac{1}{4}} \Rightarrow -8$ $\therefore P\left(-\frac{1}{\sqrt{2}}, -2\sqrt{2}\right)$ es un maximo Rel

$f\left(\frac{1}{2}\right) \Rightarrow \dfrac{2}{\frac{1}{4}} \Rightarrow 8$ $\therefore P\left(\frac{1}{\sqrt{2}}, 2\sqrt{2}\right)$ es un minimo Rel

	$-\infty$	0	$+\infty$
$\frac{2}{x^3}$	−	+	
	∩	∪	

del $]-\infty, 0[$ conc abajo

del $]0, +\infty[$ conc arriba

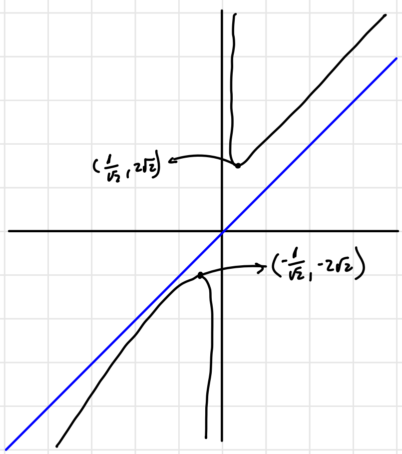
$\lim\limits_{x \to \infty} \dfrac{2x^2 + 1}{x}$ No hay A.H

$\lim\limits_{x \to \infty} \dfrac{2x^2 + 1}{x^2} \Rightarrow 2 \Rightarrow m$

$b = \lim\limits_{x \to \infty} \left(f(x) - mx\right) \Rightarrow \dfrac{2x^2 + 1}{x} - 2x \Rightarrow \dfrac{2x^2 + 1 - 2x^2}{x} \Rightarrow \dfrac{1}{x} \Rightarrow 0$

$y = 2x$

A V $\Rightarrow$ $x = 0$



$f(t) = -16t^2 + 48t + 6$

a) Verificar : $f(1) = f(2)$

$f(1)\ -16 + 48 + 6 = 38$
$f(2)\ -16 \cdot 4 + 48 \cdot 2 + 6 = 38$

b) Asumimos que $f'(t) = 0$

Dado que $f(1) = f(2)$, es continua y tambien derivable se Puede usar Roll

$f'(t) = -32t + 48 = 0$
$16(-2t + 3) = 0$
$t = 3/2$

$f(\frac{3}{2}) \Rightarrow -16 \cdot \left(\frac{3}{2}\right)^2 + 48 \cdot \frac{3}{2} + 6$

$-4 \cdot 9 + 24 \cdot 3 + 6$
$-36 + 72 + 6$
$= 42$ Pies/seg en el instante $t = 3/2$